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Team Rocket

Abstract

We predict whether Player 1 (P1) wins a Gen 2 OU Pokémon Showdown battle from P1's team, Player 2's (P2) lead and 30 turns. About 250 numeric and 1,700 sparse species/move features feed a 0.7/0.3 blend of LightGBM and a strongly regularised logistic regression. Six feature groups add game knowledge to a battle-state baseline: an estimate of P2's hidden team, win conditions, faints missing from the log, P1's items, the HP-lead trajectory and a leakage-safe "player fingerprint". The model reaches 69.3% 5-fold CV accuracy and 68.28% on the full leaderboard (top 10).

1. Introduction

We have 10,000 training battles (exactly 50% P1 wins) and 5,000 test battles. Two findings shaped the design. First, 17 training battles are exact duplicates (same timeline and P2 lead): we keep one copy, since twins split across CV folds leak labels, and reuse the known label for the 8 test battles that copy a training battle. Second, P1 averages 1.69 recorded faints per battle but P2 only 0.50, while P2's Pokémon leave at $\leq$15% HP without a recorded faint twice as often (0.28 vs. 0.12). P2 faints thus seem under-logged.

2. Evaluated Approaches

Feature engineering. The **baseline** describes the state at turn 30 for each side and as a P1−P2 difference: Pokémon fainted and alive, statuses, HP at checkpoints, move usage, type matchups (Gen 2 chart), species/move bag-of-words counts and an *estimated total HP* with unseen Pokémon at full HP, the strongest single feature. Six groups add game knowledge. **p2inf** estimates P2's unseen Pokémon from label-free team co-occurrence and compares P1's survivors with P2's expected pool. **wincon** flags boosted sweepers, Explosion, Spikes, sleep and recovery moves. **faint** targets the asymmetry above: Pokémon that left at $\leq$15% HP and never returned count as fainted in a corrected HP total. **items** counts P1's items; **traj** summarises the HP-lead curve and speed control. **fp** uses players' tendency to reuse teams: the smoothed win rate of P1's exact team and of its species set (target encoding with 5 pseudo-battles at the prior). As it uses labels, it is computed out-of-fold, and inside CV only from that fold's training data.

Table 1. Removing one group from the full model (LightGBM, 10 shared CV splits). Bars: log-loss increase ($\times 10^{-4}$).

Variant	Acc. %	Log-loss	Worse in	Δ log-loss
All six groups	68.80	0.5768	–	
− fp	68.47	0.5806	9/10	+38
− wincon	68.64	0.5800	10/10	+32
− p2inf	68.77	0.5781	8/10	+13
− faint	68.56	0.5779	8/10	+11
− items	68.72	0.5775	7/10	+7
− traj	68.79	0.5763	5/10	−5 (neutral)

Models. We compared LightGBM configurations and L_2-regularised logistic regression. Settings barely mattered (<0.004 log-loss), so we chose small randomised trees (7 leaves, 1,500 rounds, rate 0.02, 30% columns, 80% rows). Logistic regression was best at C=0.01 but errs differently from the trees, so blending helped; the LightGBM weight was searched over $0, 0.1, \ldots, 1$ (flat over 0.6–0.8).

Validation strategy. All decisions used 5-fold stratified CV repeated twice (10 splits), with every variant evaluated on the *same* splits. Log-loss, less noisy than accuracy, was the main guide, and a change had to win most splits. Added alone, only fp and wincon helped clearly, but together the six groups improved log-loss in 10/10 splits; removing any one except the neutral traj hurt (Tab. 1).

3. Results

69.3%	68.28%	Top 10
CV accuracy	full leaderboard	final ranking

The final model uses all six groups and predicts a P1 win when $0.7\,p_{\text{LightGBM}} + 0.3\,p_{\text{LogReg}} > 0.5$; LightGBM averages 5 seeds, trained on all 9,983 unique battles. CV accuracy rose from 68.1% (baseline) to 68.5–68.8% (six groups) to 69.0% (blend). As the blend weight was picked on those same predictions, we re-measured the final pipeline on 5 fresh splits: 69.3% ± 0.7, log-loss 0.571. The full leaderboard gave 68.28%; the 1.0-point gap is within noise (test-set standard error alone ≈0.65 points), so CV predicted the leaderboard well. We chose it as every component won most splits and further tuning moved scores less than this noise.

References

[1] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2nd edition, 2009.

[2] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor. *An Introduction to Statistical Learning: With Applications in Python*. Springer, 2023.

[3] I. Spinelli. Fundamentals of Data Science, lecture slides: Linear Models, 2026. Bias–variance and overfitting (pp. 30–32, 61–72); maximum likelihood, the logistic model, negative log-likelihood and its gradient (pp. 35–50); NLL and cross-entropy (pp. 48, 54–57); cross-validation and model evaluation (pp. 60, 78–88); ridge regularisation and the Gaussian prior (pp. 73–75); choosing λ (p. 77).

[4] I. Spinelli. Fundamentals of Data Science, lecture slides: Optimization, 2026. Gradient descent, step size and cost history (pp. 50–66).

[5] I. Spinelli. Fundamentals of Data Science, lecture slides: Trees and Ensembles, 2026. Decision trees, Gini index and entropy, regression trees and pruning (pp. 11–31); ensembles, weighted combination and stacking (pp. 33–39); bagging and random forests (pp. 41–44); boosting, AdaBoost, gradient boosting and XGBoost (pp. 45–52).